

Preliminary Planning Report

Smart Window Opening Control Strategy for a University Library

Prepared as an early-stage guidance document for concept design and control logic development.

Project Context	Assumption
Building type	Typical ground floor of a university library in Germany
Outdoor environment	Quiet campus setting, no significant vehicle traffic
Summer operation	No mechanical cooling assumed
Winter operation	Heating available
Main control idea	Occupancy determines basic ventilation; CO2, temperature and noise provide corrections

Key recommendation

Use the number of occupants as the main feed-forward signal. Use CO2 as feedback to correct the estimate. Use temperature difference to avoid unwanted heat gain or heat loss. Use noise monitoring as a protection rule, not as the main control driver.

1. Purpose and Scope

This report provides a simplified planning framework for controlling the opening angle and number of opened windows in a smart university library. The objective is to maintain comfortable temperature, acceptable air quality and a quiet reading environment while avoiding unnecessary winter heat loss.

The document is intended for early-stage planning. It is not a final HVAC design calculation and should be calibrated later using real building measurements.

2. Basic Planning Assumptions

Item	Symbol	Assumed value
Floor area	Af	1,000 m ²
Room height	Hr	3.5 m
Indoor air volume	V	3,500 m ³
Typical maximum occupancy	Nmax	150-200 people
Operable windows	Nw	40 windows
Maximum effective opening area per window	Amax	0.8 m ²
Winter comfort temperature	Tin,winter	20-22 deg C
Summer comfort temperature	Tin,summer	22-26 deg C
Target CO ₂ concentration	CO _{2,target}	Below 1,000 ppm
Preferred indoor noise level	Lin,max	Below 40 dB(A)

The numerical assumptions should be replaced by actual project data during the detailed design stage. The control logic, however, can remain the same.

3. Main Control Philosophy

The system should not wait until indoor CO₂ becomes high. CO₂ is a delayed signal. A better approach is to estimate the required ventilation from the number of people first, then adjust it using measured CO₂ and temperature data.

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Final window level = Occupancy-based level
+ CO2 correction
+ temperature correction
+ noise correction
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In this project, the campus environment is assumed to be quiet. Therefore, noise is treated as a protective constraint rather than the dominant control factor.

4. Occupancy-Based Ventilation Demand

The number of occupants is the main driver of fresh air demand. For preliminary planning, use 9 L/s per person as a simple fresh air estimate.

$$Q_{occ} = N \times q_p$$

Where:

Q_{occ} = required outdoor airflow (m³/s)

N = number of occupants

q_p = outdoor airflow per person = 0.009 m³/s/person

Number of occupants	Required airflow Q _{occ}	Equivalent airflow	Air changes per hour for V=3500 m ³
20	0.18 m ³ /s	648 m ³ /h	0.19 h ⁻¹
50	0.45 m ³ /s	1,620 m ³ /h	0.46 h ⁻¹
100	0.90 m ³ /s	3,240 m ³ /h	0.93 h ⁻¹
150	1.35 m ³ /s	4,860 m ³ /h	1.39 h ⁻¹
200	1.80 m ³ /s	6,480 m ³ /h	1.85 h ⁻¹

5. Simplified Window Ventilation Model

Natural ventilation through windows can be estimated using a simple opening-flow relationship. This is suitable for early planning, although detailed design should use measured wind conditions or simulation.

$$Q = C_d \times A_{eff} \times \sqrt{2 \times \Delta P / \rho}$$

Where:

Q = airflow rate (m³/s)

C_d = discharge coefficient, use 0.6 for planning

A_{eff} = total effective opening area (m²)

ΔP = pressure difference from wind and stack effect (Pa)

ρ = air density, about 1.2 kg/m³

For a first estimate, use ΔP = 1.0 Pa. This gives approximately:

$$Q / A_{eff} = 0.775 \text{ m}^3/\text{s per m}^2$$

$$A_{eff, required} = Q_{occ} / 0.775$$

Occupants	Q _{occ}	Required effective opening area
20	0.18 m ³ /s	0.23 m ²
50	0.45 m ³ /s	0.58 m ²
100	0.90 m ³ /s	1.16 m ²
150	1.35 m ³ /s	1.74 m ²
200	1.80 m ³ /s	2.32 m ²

6. Window Opening Angle

The effective opening area of one window can be estimated from its opening angle. For this planning model, the relationship is simplified as follows:

$$A_{eff,1} = A_{max} \times \sin(\alpha)$$

Where:

A_{eff,1} = effective opening area of one window

A_{max} = maximum effective opening area of one window, assumed 0.8 m²

α = window opening angle

Opening angle	Effective area per window
5 deg	0.07 m ²
10 deg	0.14 m ²
15 deg	0.21 m ²
20 deg	0.27 m ²
30 deg	0.40 m ²

Opening angle	Effective area per window
45 deg	0.57 m2

For a library, the recommended strategy is many windows with small opening angles. This reduces draught, avoids local discomfort and keeps the reading environment calm.

7. Recommended Window Opening Levels

The following table can be used as the initial control schedule for normal operation in spring, summer and autumn when outdoor temperature is suitable.

Level	Occupancy	Recommended window action	Planning comment
0	0-20	Closed, or 2 windows at 5 deg	Minimum ventilation; quiet and energy saving
1	20-50	4 windows at 10 deg	Basic ventilation
2	50-100	6 windows at 15 deg	Moderate occupancy
3	100-150	8 windows at 15 deg, or 6 windows at 20 deg	High occupancy
4	150-200	10 windows at 15 deg, or 8 windows at 20 deg	Near full occupancy
5	>200 or high CO2	12-16 windows at 20 deg	Strong ventilation; consider mechanical support

8. CO2 Feedback Correction

Occupancy estimates can be wrong, and actual natural ventilation depends on wind, temperature difference and window position. Therefore, measured CO2 should correct the occupancy-based level.

Indoor CO2 concentration	Correction to window level
Below 700 ppm	Reduce by one level if temperature is comfortable
700-900 ppm	Keep the occupancy-based level
900-1,000 ppm	Increase by one level
1,000-1,500 ppm	Increase by two levels or use short pulse ventilation
Above 1,500 ppm	Use Level 5 strong ventilation
Above 2,000 ppm	Warning or facility manager intervention

In normal operation, the system should aim to keep CO2 below approximately 1,000 ppm.

9. Thermal Comfort Rules

9.1 Summer and transition seasons

In summer, the library is assumed to have no mechanical cooling. Natural ventilation is useful only when outdoor air is cooler than indoor air. If outdoor air is warmer, window opening should be limited to the amount needed for air quality.

Condition	Recommended action
Outdoor temperature lower than indoor temperature	Ventilation can be increased, especially if occupancy or CO2 is high
Outdoor temperature higher than indoor temperature	Avoid long-term window opening; ventilate only for CO2 control
Early morning or night air is cooler	Use night or morning ventilation to pre-cool the building
Indoor temperature above 26 deg C	Increase ventilation only if outdoor air is cooler

9.2 Winter

In winter, opening windows causes heating energy loss. The heat loss can be estimated as:

$$Q_{\text{loss}} = 1200 \times Q_v \times (T_{\text{in}} - T_{\text{out}})$$

Where:

Q_{loss} = ventilation heat loss (W)

Q_v = ventilation airflow (m³/s)

T_{in} = indoor temperature (deg C)

T_{out} = outdoor temperature (deg C)

Example: if $Q_v = 1.0$ m³/s, $T_{\text{in}} = 21$ deg C and $T_{\text{out}} = 5$ deg C, then $Q_{\text{loss}} = 19,200$ W. This is a significant heating load, so winter ventilation should mainly use short pulse ventilation rather than long continuous opening.

Occupancy	Recommended winter action
Below 50 people	Keep windows mostly closed; ventilate only when CO2 increases
50-100 people	Short ventilation periods when CO2 approaches 1,000 ppm
100-150 people	Pulse ventilation every 20-30 minutes if needed
More than 150 people	More frequent pulse ventilation or mechanical assistance

10. Acoustic Control

The outdoor environment is assumed to be a quiet university campus without significant road traffic. Therefore, outdoor noise is not expected to be the main limiting factor. However, indoor acoustic comfort remains important.

Recommended indoor noise target:

$$L_{\text{in}} \leq 40 \text{ dB(A)}$$

Condition	Recommended action
Indoor noise below 40 dB(A)	Normal occupancy-based control
Indoor noise above 40 dB(A)	Reduce opening by one level
Temporary outdoor noise event	Close windows facing the noise source
Low occupancy and very quiet reading area	Prefer smaller opening angles

11. Simplified Control Algorithm

Every 1-5 minutes:

1. Read inputs:

N , T_{in} , T_{out} , CO_2 , L_{in} , L_{out} , season

2. Set base level from occupancy:

0-20 people -> Level 0

20-50 people -> Level 1

50-100 people -> Level 2

100-150 people -> Level 3

150-200 people -> Level 4

more than 200 -> Level 5

3. Apply CO₂ correction.

4. Apply temperature correction.

5. Apply noise correction.

6. Limit final level between 0 and 5.

7. Send command to window actuators.

To avoid unstable window movement, the system should include a minimum hold time. For example, after changing level, the system should wait at least 5-10 minutes before making another non-urgent adjustment.

12. Recommended Sensor System

Sensor type	Purpose
Occupancy or people-counting system	Main input for ventilation demand
Indoor CO2 sensors	Air quality feedback
Indoor temperature sensors	Thermal comfort feedback
Outdoor temperature sensor	Determines whether ventilation helps or hurts thermal comfort
Indoor noise sensors	Acoustic comfort protection
Outdoor noise sensor	Detection of temporary outdoor noise events
Window position sensors	Confirm actual window opening state

CO2 sensors should be installed in occupied reading zones, not directly beside windows. Temperature sensors should be placed both near windows and in central zones because comfort conditions may differ.

13. Commissioning and Calibration

The proposed values should be treated as initial settings. After installation, the building should be observed for at least several weeks under different occupancy and weather conditions.

Check whether CO2 remains below the target level during peak occupancy.

Check whether users report draught or cold discomfort near windows.

Check whether winter pulse ventilation causes excessive temperature drops.

Check whether indoor noise remains acceptable during normal campus activity.

Adjust each occupancy level based on measured CO2 response and user feedback.

14. Summary of Key Recommendations

Use occupancy as the main feed-forward input for window control.

Use CO2 as feedback to correct the occupancy-based ventilation estimate.

Prefer many windows with small opening angles instead of a few windows opened widely.

In summer, increase ventilation only when outdoor air is cooler than indoor air, except when CO2 control requires fresh air.

In winter, avoid long continuous opening and use short pulse ventilation instead.

Treat noise as a protective constraint because the campus outdoor environment is generally quiet.

Calibrate the control table after installation using measured CO2, temperature and comfort feedback.

The simplest final control principle is:

Occupancy determines the basic window level. CO2 corrects it. Temperature decides whether to increase or limit ventilation. Noise protects the quiet reading environment.

Appendix: Quick Reference Table

Occupants	Suggested action	Main purpose
0-20	Closed, or 2 windows at 5 deg	Quiet and energy saving
20-50	4 windows at 10 deg	Basic ventilation

Occupants	Suggested action	Main purpose
50-100	6 windows at 15 deg	Moderate ventilation
100-150	8 windows at 15 deg	High occupancy ventilation
150-200	10 windows at 15 deg or 8 windows at 20 deg	Near full occupancy
>200 or CO2 high	12-16 windows at 20 deg	Strong ventilation